

VINAY KUMAR REDDY SIRIGIREDDY

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PROFESSIONAL SUMMARY

MS Aerospace Systems Engineering graduate (Wright State, GPA 3.7, May 2026) with two years of hands-on system safety engineering experience on a safety-critical electrified aircraft program. Currently performing FHA, FMEA, FMECA, and fault propagation analysis for a fault-tolerant propulsion architecture using ARP 4761 and MIL-STD-882 safety assessment methodology. Built an MBSE-integrated safety verification framework that automatically evaluates compliance against airworthiness constraints — no manual handoff between the analytical model and the requirement. Experienced working across propulsion, structures, electrical, and software domains simultaneously to identify hazards before they reach hardware.

EDUCATION

Master of Science, Aerospace Systems Engineering

Wright State University, Dayton, OH

GPA: 3.7

May 2026

- Thesis: MATLAB-MBSE Integrated Sizing Framework for eVTOL Aircraft: Evaluating Peak Power Demands on MTOW Convergence — Advisor: Dr. Darryl K. Ahner (Accepted April 2026)
- Applied FHA, FMEA/FMECA, and fault propagation analysis across full system lifecycle. Automated safety constraint verification against FAA Part 23 Amendment 64 inside SysML parametric model — validated within 2% of three industry aircraft.

Bachelor of Science, Aeronautical Engineering

Bharath Institute of Higher Education and Research, India

GPA: 3.5

May 2023

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Wright State University, Dayton, OH

Feb 2024 – Present

- **FHA / FMEA / FMECA & Hazard Analysis (current):** Performing Functional Hazard Assessment (FHA), Failure Modes Effects and Criticality Analysis (FMECA), and fault propagation analysis for fault-tolerant LPC/TR eVTOL propulsion architecture per ARP 4761 and MIL-STD-882 methodology. Identifying single-point failure modes across motor, inverter, and battery subsystems, assessing severity and criticality, and defining containment strategies.
- **Safety Requirement Development & Verification:** Defined airworthiness safety requirements from FAA 14 CFR Part 23 Amendment 64 and implemented them as executable Boolean constraints in SysML. Automated pass/fail evaluation of safety targets — MTOW limits, minimum SOC floor, hover thrust margins, peak power discharge constraints — across 17 design configurations without manual interpretation.
- **Cross-Functional Safety Coordination:** Coordinated safety analysis across propulsion, battery, structural, and aerodynamic domains simultaneously. Tracked cross-system hazard dependencies, resolved conflicting safety constraints between subsystems, and confirmed integrated system compliance in a single simulation pass — working across four engineering disciplines without a formal team structure.
- **Hazard Identification & Risk Quantification:** Identified energy-only battery sizing as a critical safety risk: the design produced an aircraft physically incapable of hover — a Category I hazard. Quantified the gap (1,247 kg MTOW underestimate), implemented dual-constraint formulation to resolve it, and documented the corrective action with full traceability back to the originating safety requirement.
- **Safety Assessment Documentation:** Documented complete safety assessment architecture in SysML: requirements diagram, constraint blocks, parametric bindings, and automated verification logic. Created reusable process templates for requirement capture, hazard traceability, and instance-based safety evaluation — equivalent to System Safety Program Plan (SSPP) artifact set for a research program.
- **V&V & Technical Reporting:** Validated safety framework outputs within 2% of Joby S4, Wisk Cora, and BETA ALIA-250 published specifications. Prepared thesis manuscript, technical presentations, and design review materials documenting safety methodology, hazard analysis results, and regulatory compliance status.

Advanced Rocket Propulsion Intern | MARS Exploration Pvt. Ltd.

May – Dec 2021

- Evaluated Isp and combustion safety performance for ion thruster and hybrid propulsion concepts using NASA CEA. Identified optimal O/F ratio range with ~12% efficiency improvement while assessing combustion stability boundaries.

STK Mission Analysis Intern | SS Technologies

Aug 2020 – Jan 2021

- Modeled satellite orbits and aircraft trajectories in AGI STK, evaluating mission safety performance under environmental and operational constraints.
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SELECTED PROJECTS

SysML Fault-Propagation Architecture — Hybrid LPC/TR eVTOL | CATIA Magic / Cameo2024 – Present

- 10-diagram SysML model (BDD, IBD, STM, ACT, PAR, REQ, UC) performing FMECA across propulsion, battery, and flight control subsystems. Defining failure containment boundaries and safety interface requirements per ARP 4761 FHA methodology.

eVTOL Aircraft Sizing Web Application | React / GitHub Pages — Dr. Ahner 2024 – 2025

- Full-stack sizing tool with 23 analysis tabs including safety constraint checking, Monte Carlo trade studies, aeroacoustic noise model, and PDF export. Live at evtolsizer.live.

CFD — Hoverbike Propulsion Systems | Cradle CFD / SOLIDWORKS 2023

- Evaluated aerodynamic interference and thrust-to-power tradeoffs for overlapping propeller configurations for AAM applications.

FEA Structural Optimization — Hoverbike Ducts | HyperMesh / ANSYS 2023

- Structural safety analysis comparing carbon fiber composites vs. 6-series aluminum under flight loads. Met all safety factor requirements at reduced mass.

20-Seater Business Jet Design (Capstone) | OpenVSP / SOLIDWORKS 2022 – 2023

- Conceptual sizing, configuration layout, and preliminary structural design benchmarked against Gulfstream G500.

TECHNICAL SKILLS

System Safety Engineering: FHA, FMEA, FMECA, FTA, CCA/CMA, hazard identification and risk quantification, safety requirement development, safety constraint verification, System Safety Program Plan (SSPP), ARP 4761, MIL-STD-882, ARP 4754A, FAA Part 23 Amendment 64, DO-178C / DO-254 (awareness)

Systems Engineering & MBSE: SysML (BDD, IBD, PAR, REQ, STM, ACT, UC), CATIA Magic / Cameo Systems Modeler, requirements traceability, architecture decomposition, V&V planning, digital thread, cross-functional coordination

Analysis & Simulation: MATLAB / Simulink, OpenVSP, ANSYS, Cradle CFD, HyperMesh, NASA CEA, STK, XFOIL, Python

CAD & Design: SOLIDWORKS (CSWP, CSWA), CATIA V5, AutoCAD, Fusion 360

Programming: MATLAB, Python (data science, scripting, automation), JavaScript / React

Aerospace Domains: Safety-critical aircraft systems, eVTOL / electrified aircraft, distributed electric propulsion, AAM, aircraft performance, aerodynamics, propulsion, FEA

CERTIFICATIONS

- SOLIDWORKS: CSWP, Drawing Tools Professional, Additive Manufacturing Associate, CSWA
- Dassault Systemes: 3D Swym Associate | IBM: Python for Data Science, AI & Development
- NPTEL: Aircraft Performance; Manufacturing Process Technology I & II | MathWorks: MATLAB
- Udemy: Product Development & Systems Engineering | Verzeo: AutoCAD
- Publication: Hoverbike aerodynamic and propulsion analysis — IJISRT

ACTIVITIES & LANGUAGES

Professional Membership: ASME **Languages:** English, Hindi, Telugu, Tamil